Proyecto del Plano Inclinado

Determinación teórica y experimental del tiempo de descenso de un bloque con fricción cinética. Reporte generado el 10/10/2026.

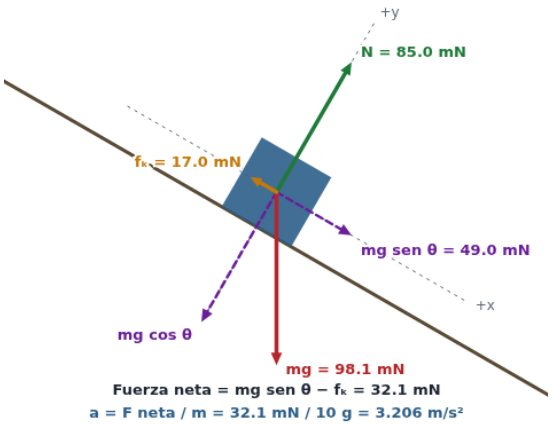
1. Parámetros

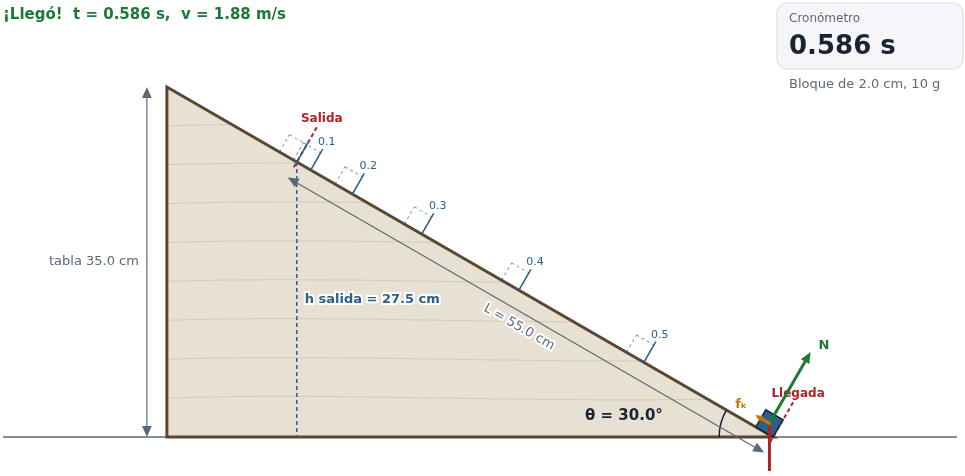
Objeto	Tamaño	m	θ	h salida	L recorrida	μ_k	μ_s	g	v_0
bloque	2.0 cm	10 g	30°	27.5 cm (tabla 35.0 cm)	55.0 cm	0.200	0.25	9.81 m/s²	0 m/s

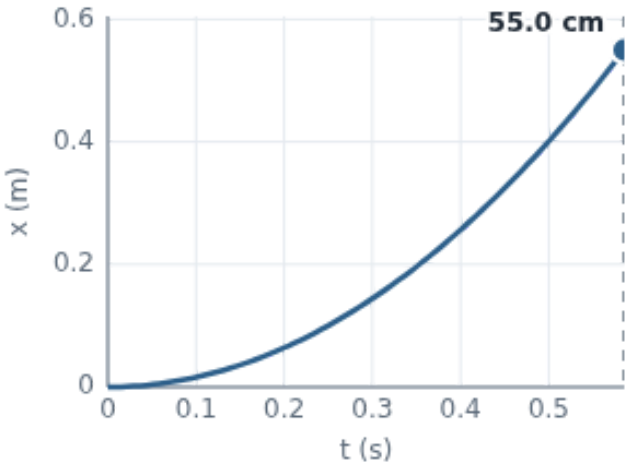
2. Análisis de fuerzas y cálculo teórico

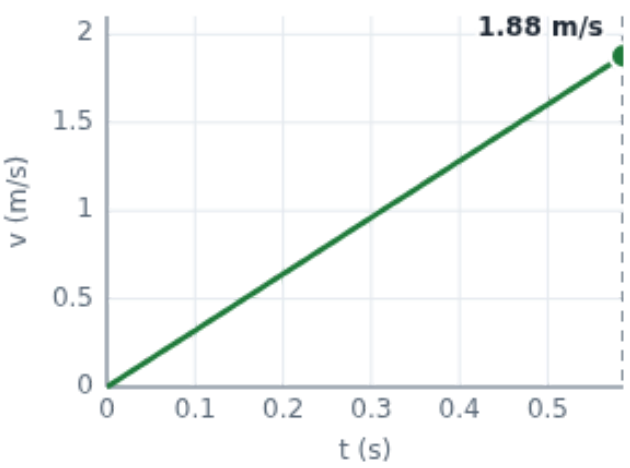
N = mg cos θ; ma = mg sen θ − μ_k mg cos θ ⟹ a = g(sen θ − μ_k cos θ) = 9.81(0.5000 − 0.1732) = **3.206 m/s²**

Recorrido L = 55.0 cm (altura de salida h = L·sen θ = 27.5 cm); t = √(2L/a) = √(2 × 0.550 / 3.206) = **0.586 s**; v = a·t = **1.88 m/s**

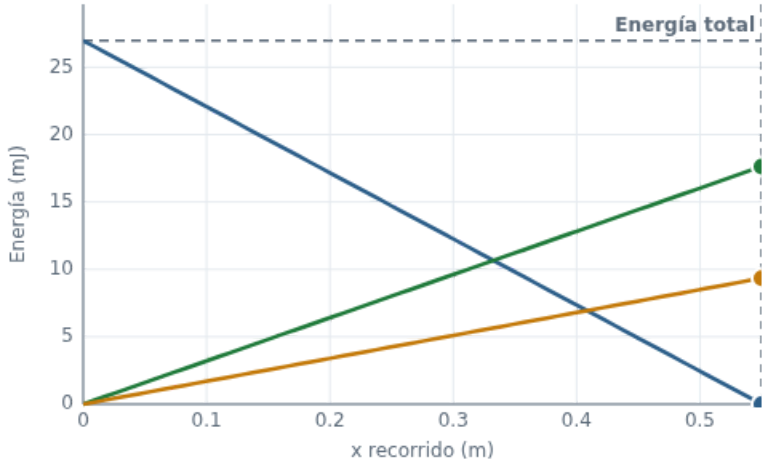





55.0 cm
x (m)
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
t (s)

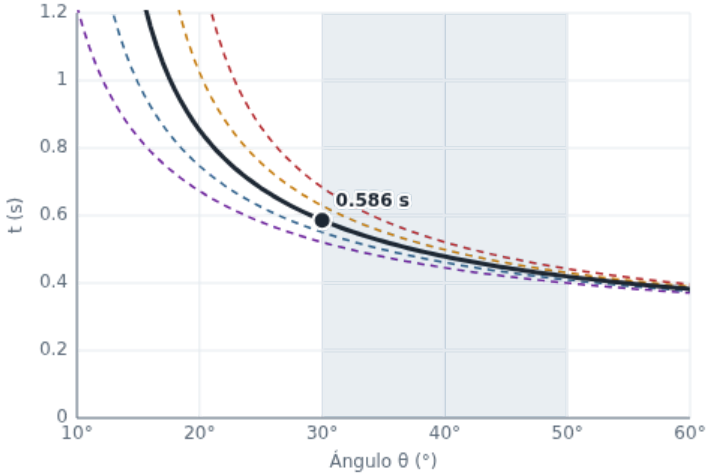

1.88 m/s
v (m/s)
2
1.5
1
0.5
0
0
0.1
0.2
0.3
0.4
0.5
t (s)

3. Energía



4. Sensibilidad (L = 55.0 cm)

θ	μ_k = 0.10	μ_k = 0.15	μ_k = 0.20	μ_k = 0.25	μ_k = 0.30
30°	0.521 s	0.550 s	**0.586 s**	0.629 s	0.683 s
35°	0.478 s	0.499 s	0.523 s	0.551 s	0.585 s
40°	0.445 s	0.461 s	0.479 s	0.498 s	0.521 s
45°	0.420 s	0.432 s	0.445 s	0.460 s	0.476 s
50°	0.400 s	0.409 s	0.419 s	0.430 s	0.442 s



5. Registro experimental

Ensayo	Tiempo medido (s)
1	2.050
2	1.940
3	2.100
4	1.980
5	2.030
Promedio	**2.020**

Error (%) = $|t_{teórico} - t_{experimental}| / t_{experimental} \times 100 = |0.586 - 2.020| / 2.020 \times 100 =$ **71.0 %**

Desviación estándar: 0.062 s. Nota según la escala: **90**. Aceleración medida: 0.270 m/s². μ_k experimental: 0.546.

